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그래프 신경망 설명을 위한 양극화된 엣지 점수  
분포 학습

Beyond Graph Information Bottleneck: Creating Bimodal Edge  
Score Distributions for GNN Explanation

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# Beyond Graph Information Bottleneck: Creating Bimodal Edge Score Distributions for GNN Explanation

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The study was conducted in accordance with Code of Research Ethics<sup>1</sup>.

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<sup>1</sup> Declaration of Ethical Conduct in Research: I, as a graduate student of Korea Advanced Institute of Science and Technology, hereby declare that I have not committed any act that may damage the credibility of my research. This includes, but is not limited to, falsification, thesis written by someone else, distortion of research findings, and plagiarism. I confirm that my thesis contains honest conclusions based on my own careful research under the guidance of my advisor.



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### 초 록

그래프 신경망의 사후 설명은 예측 결과에 가장 큰 영향을 미친 부분 그래프를 식별함으로써 모델의 의사 결정 과정을 해석하는 것을 목표로 한다. 그러나 이산적인 부분 그래프는 미분이 불가능하기 때문에, 기존 설명 기법들은 연속적인 엣지 중요도 점수를 출력하며, 이로 인해 경계가 불분명하고 혼합된 점수 분포가 형성되는 경우가 많다. 또한 학습 과정에서는 그래프 신경망의 예측 결과만 제공될 뿐, 엣지 수준의 정답 레이블은 존재하지 않는다는 한계가 있다. 본 연구에서는 약한 예측 보존 신호를 명시적인 엣지 수준 지도 정보로 변환하는 기법을 제안한다. 제안한 방법은 설명 정확도를 향상시키는 동시에, 엣지 점수를 중요한 엣지와 중요하지 않은 엣지로 뚜렷하게 양극화한다. 또한 제안한 방법을 기존 최신 설명 기법에 적용할 경우, 엣지 점수 분포를 더욱 명확하게 형성하여 설명 부분 그래프의 분리 가능성을 크게 향상시킴을 보인다. 네 가지 벤치마크 데이터셋에 대한 실험 결과, 엣지 수준 설명 정확도가 일관되게 향상되었다.

핵심 낱말 그래프 신경망, 설명 가능한 인공지능, 그래프 정보 병목

### Abstract

Post-hoc explanation for graph neural networks (GNNs) aims to interpret their predictions by identifying the subgraphs most responsible for a decision. Since discrete subgraphs are not differentiable, existing explainers produce continuous edge-importance scores as output, often yielding blurry, mixed score distributions. The main challenge is the absence of edge-level ground-truth labels, as only the predictions of GNNs are provided as training data. We convert weak prediction-preserving signals into explicit edge supervision during training. As a result, it markedly improves explanation accuracy while sharply polarizing edges into important versus unimportant ones. We demonstrate that it creates improved shapes of score distributions when applied to state-of-the-art explainers, resulting in more separable subgraphs. Experiments on four benchmarks show that it improves the accuracy of edge-level explanations of state-of-the-art baselines.

Keywords Graph neural networks, Interpretable AI, Graph information bottleneck

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# Chapter 1. Introduction

Graph neural networks (GNNs) have emerged as a powerful model for learning over graph-structured data [13, 29, 35, 33], achieving remarkable success across a wide range of domains [7, 44]. Despite active work on improving the accuracy of GNNs, their reasoning process remains unclear, hindering their deployment in settings where trust and accountability are essential [38, 40, 1, 22].

The *post-hoc explanation* of GNNs seeks to reveal substructures of a graph, such as nodes, edges, or motifs, that serve as the main evidence for a model’s prediction. Early methods such as GNNExplainer [36] and PGExplainer [21] model explanation as sampling a sparse edge mask that maximizes the mutual information with the original prediction, while recent approaches leverage causal graphs [30], counterfactual explanations [20] or generative modeling [3, 16] to provide more expressive explanations.

In practice, these post-hoc explainers are trained under indirect supervision without access to ground-truth edge-level annotations. Most existing methods [39, 36, 21, 31, 34, 43] adopt the information bottleneck as their objective function [26, 32], which aims to identify a sparse subgraph that preserves the original prediction when provided to the GNN. Typically, the explainer assigns a single continuous score to each edge to indicate its importance. To facilitate the separation between important and unimportant edges in the output, a mask entropy regularization term is typically introduced to encourage the scores to be pushed toward 0 or 1 [36, 21].

However, even when these methods achieve high accuracy in explanation benchmark, the resulting importance scores often fail to exhibit intuitive binarization. Figure 1.1a shows a case where the edge scores are highly ambiguous and poorly polarized regardless of the AUC score. Consequently, it becomes difficult for practitioners to clearly know which edges the explainer considers important [9]. A straightforward solution is to strengthen the entropy regularization to enforce sharper score distributions. However, as shown in Figure 1.1b, naively increasing the regularization strength does not effectively address the problem and instead leads to over-confident predictions. Note that the colors in the figure refer to ground-truth information that is unavailable in practical usage.

This issue is critical because the essential objective of post-hoc explanation is not to preserve raw continuous scores themselves, but to extract a discrete explanatory subgraph from them. In practice, users typically obtain explanations by thresholding edge scores or selecting the top- $k$  edges. However, existing methods provide no clear criterion for choosing an appropriate threshold, and top- $k$  extraction depends only on relative ranking rather than the actual separability of the scores. As a result, even highly ambiguous score distributions can still achieve strong explanation AUC despite lacking a clear distinction between important and unimportant edges. Moreover, in practical scenarios, the appropriate value of  $k$  is generally unknown because the size of the true explanatory subgraph is unavailable and can vary significantly across graphs.

These limitations highlight that explanation quality should not be evaluated solely by ranking accuracy [2, 6], but also by whether the explainer produces naturally polarized and clearly binarized score distributions that support reliable and interpretable subgraph extraction. Therefore, without strong bimodality and clear binarization in the score distribution, it is difficult to regard a GNN explainer as practical or trustworthy.

As a solution, we propose BEEP (Bimodal Edge Explanation via Polarization), which is a general self-guided framework for enhancing the explanation quality of already trained explainers. BEEP does

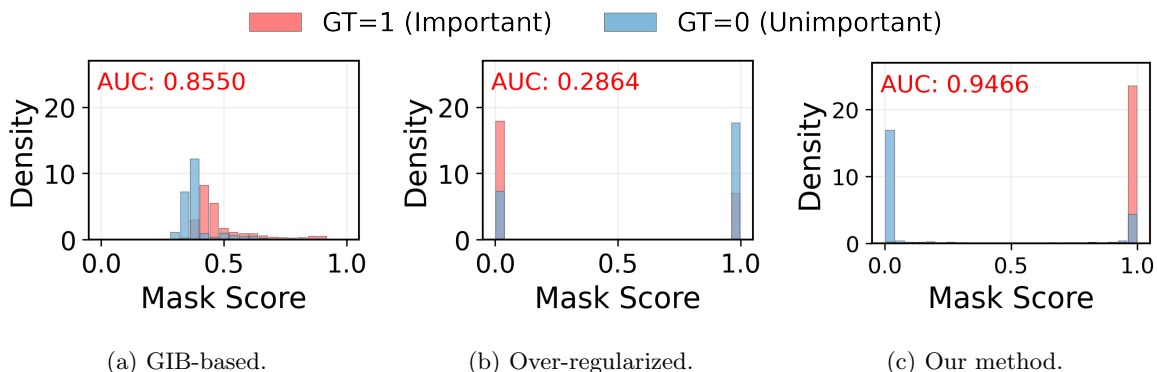


Figure 1.1: Edge score distributions on BA3. (a) GIB-based explainer creates a blurry distribution despite rationally high AUC. (b) An over-confident distribution is created when the entropy regularizer is overly emphasized for binarization. (c) BEEP creates a desirable distribution, making clear bimodality near 0 and 1 with high accuracy.

not require modifying or retraining an existing explainer; it extracts high-confidence signals from the explainer and uses them to train a guided explainer with self-supervision. Unlike previous methods that rely on unsupervised mutual information, BEEP provides clear guidance to the guided explainer by propagating pseudo-labels derived from the base explainer. This approach results in a clear, binarized score distribution that separates important and unimportant edges even without complicated regularization.

Our experiments on four benchmark datasets, including synthetic and real-world molecular graphs, show that BEEP consistently improves the AUC and F1 scores of state-of-the-art post-hoc explainers. Beyond accuracy, BEEP yields markedly more interpretable score shapes: binarization score and bimodality increase across datasets, indicating clearer extraction of the important edges. Ablation studies show that these gains are robust to the choice of a base explainer and persist under round-wise self-guidance.

Our contributions are summarized as follows:

- **Theoretical Analysis.** We provide a theoretical analysis explaining why GIB fails to achieve effective binarization, even when a strong regularizer is applied, whereas our approach succeeds.
- **Pseudo-labeling Strategy.** We propose a pseudo-labeling strategy that converts soft edge scores into reliable discrete signals. Our pseudo-labels systematically push edge scores away from mid-range values, driving them toward stronger binarization.
- **Extensive Experiments.** We conduct comprehensive experiments on both synthetic and real-world datasets. Our method consistently improves both predictive accuracy and distributional interpretability over existing baselines, while remaining robust across a wide range of ablation settings and maintaining computational efficiency.

## Chapter 2. Preliminaries and Related Work

**Graph Classification.** Let  $G = (\mathbf{X}, \mathbf{A})$  be an undirected graph, where  $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$  is the node feature matrix,  $\mathbf{A} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{V}|}$  is the adjacency matrix,  $\mathcal{V}$  is the set of nodes, and  $d$  is the feature size. For  $\mathbf{A}$ , each element is  $a_{uv} = 1$  if nodes  $u$  and  $v$  are connected and 0 otherwise. The edge set is  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ , where  $(u, v) \in \mathcal{E}$  if and only if  $a_{uv} = 1$ . In the *graph classification* setting, each graph  $G$  is associated with a label  $Y \in \{1, \dots, C\}$  that describes its global structural or semantic property, where  $C$  is the number of classes. Given a set  $\mathcal{G}$  of training graphs, we assume to have a GNN classifier  $f$  trained on  $\mathcal{G}$  which is our main target of explanation.

**Post-hoc Explanation of GNNs.** Given a trained GNN  $f$ , *post-hoc explanation* aims to produce a subgraph  $G^*$  that most significantly contributes to the GNN’s prediction  $\hat{Y} = f(G)$ . Although the goal of explanation is to find a discrete subgraph, most existing explainers return continuous edge importances  $\hat{\mathbf{S}} = h(f, G) \in [0, 1]^{|\mathcal{V}| \times |\mathcal{V}|}$ , where  $h$  denotes an explainer model. Since the problem is unsupervised, the evaluation of the generated subgraph is typically done by comparing it with the ground-truth edge labels  $\mathbf{S} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{V}|}$ , which represent the importance of each edge for the label  $Y$  of graph  $G$ . Such ground-truth is only for evaluation and is not provided at training time.

**Graph Information Bottleneck (GIB).** With the GIB principle [32], the ideal explanation of GNN  $f$  satisfies

$$\max_{G^*} \text{MI}(\hat{Y}, G^*) = \max_{G^*} (H(\hat{Y}) - H(\hat{Y} \mid G^*)), \quad (2.1)$$

where  $H$  is the entropy function. Since  $H(\hat{Y})$  is fixed by the given  $f$ , maximizing  $\text{MI}(\hat{Y}, G^*)$  is equivalent to minimizing  $H(\hat{Y} \mid G^*)$ , which makes the extracted subgraph  $G^*$  have the same class information as in the original graph  $G$ . To enforce the compactness of explanation with sparse  $G^*$ , prior works typically include a regularizer term for its size [21, 31, 43, 41, 4]. This constraint encourages the explainer to focus on a minimal set of salient edges.

**Related Work.** The principle of explanation through GIB has been consistently adopted across a wide spectrum of works, from early graph mask learning in GNNExplainer [36] to parametric prediction with PGExplainer [21] and fine-tuning strategies in ReFine [31]. To address distribution shifts and improve generalization, later GIB-based methods such as MixupExplainer [43] and ProxyExplainer [4] introduced graph mixing and generation. More recently, there have been explorations into self-explainable temporal models [27], factorized explainers [10], and regression-based variants [41]. These developments highlight the influence of the GIB-based paradigm in shaping the field of GNN explanation.

Beyond the standard GIB framework, recent research explores diverse directions: SE-SGformer [15] identifies neighbor patterns for link prediction, while FORGE [28] captures multi-node interactions via cell complexes. Geometric properties like curvature have been leveraged to enhance robustness [17], and MAGE [37] synthesizes motif-based explanations to ensure structural validity.

Despite these advancements, a limitation remains in achieving an ideal distribution of importance scores. Methods relying on continuous probability distributions often yield ambiguous scores, failing to provide a definitive distinction between important and unimportant edges. Conversely, approaches that enforce strict discretization risk being overly constrained. This highlights the necessity for a method that achieves sharp binarized distinctions while retaining the representational flexibility of probabilistic modeling.



### Chapter 3. Why GIB Fails at Binarization

The binarization of edge importance scores might be achieved by adding a suitable regularizer to the GIB objective function, not only by our self-guided framework. Previous work [21] proposed an objective function that balances prediction consistency, sparsity, and the binarization of edge importance scores as follows:

$$\mathcal{L}_b(h^{(b)}) = \text{CE}(\hat{Y}, \hat{Y}^*) + \lambda_1 \sum_{(u,v) \in \mathcal{E}} \hat{s}_{uv}^{(b)} + \lambda_2 \sum_{(u,v) \in \mathcal{E}} H(\hat{s}_{uv}^{(b)}), \quad (3.1)$$

where  $H$  is the entropy function minimized if the scalar input is 0 or 1,  $\hat{Y}^* = f(G^*)$  is the prediction of the GNN  $f$  for the extracted subgraph  $G^* = (\mathbf{X}, \hat{\mathbf{S}}^{(b)})$ , and  $\hat{s}_{uv}^{(b)} \in [0, 1]$  is the soft importance score for edge  $(u, v)$  produced by the explainer  $h^{(b)}$ . The first term measures consistency with the original GNN prediction, the second term encourages score sparsity, and the third term promotes binarization, where the coefficients  $\lambda_1$  and  $\lambda_2$  make a balance.

The main limitation of Eq. (3.1) is that it has a provable tendency to converge to non-binary scores in the middle of 0 and 1, not being sufficiently polarized, as formalized in Lemma 3.1. The lemma states that there can be an upper bound on binarization that can be achieved with GIB, even with a suitable regularizer.

**Lemma 3.1** (Failure of GIB). *For the regularized GIB loss  $\mathcal{L}_b$  in Eq. (3.1), consider a stationary point where the fidelity gradient is zero:*

$$g_{uv}(\hat{\mathbf{S}}^{(b)}) := \partial \text{CE}(\hat{Y}, \hat{Y}^*) / \partial \hat{s}_{uv}^{(b)} = 0. \quad (3.2)$$

*At this point, the optimal score  $\hat{s}_{uv}^{(b)}$  converges to a fractional value independent of the edge properties:*

$$\hat{s}_{uv}^{(b)*} = \sigma(\lambda_1/\lambda_2) \in (0, 1), \quad (3.3)$$

*where  $\sigma$  is the logistic sigmoid function.*

*Proof.* Differentiating Eq. (3.1) with respect to  $\hat{s}_{uv}^{(b)}$  yields the gradient:  $g_{uv}(\hat{\mathbf{S}}^{(b)}) + \lambda_1 + \lambda_2 \log((1 - \hat{s}_{uv}^{(b)})/\hat{s}_{uv}^{(b)})$ . When the masked prediction successfully matches the original prediction, the fidelity gradient diminishes, i.e.,  $g_{uv}(\hat{\mathbf{S}}^{(b)}) \approx 0$ . Setting the remaining edge-agnostic regularization terms to zero to find the stationary point gives  $0 = \lambda_1 + \lambda_2 \log((1 - \hat{s}_{uv}^{(b)})/\hat{s}_{uv}^{(b)})$ . Solving for  $\hat{s}_{uv}^{(b)}$  directly yields the fractional fixed point  $\sigma(\lambda_1/\lambda_2) \in (0, 1)$ . Details are provided in Appendix A.1.1.  $\square$

Lemma 3.1 highlights a core limitation of GIB objectives. Once the masked subgraph prediction matches the original prediction, the fidelity gradient  $g_{uv}$  shrinks for many edges, so the optimization is dominated by the *per-edge* size and entropy regularizers. Since these regularizers are edge-agnostic, they impose a nearly uniform bias that pulls scores toward the same interior fixed point  $\sigma(\lambda_1/\lambda_2)$ , rather than splitting them into two modes. As a result, the resulting distributions may remain blurry and mixed despite achieving high AUC (see Figure 1.1a); that is, while edges are ranked correctly, they are not polarized into a clear bimodal distribution.

A seemingly simple fix is to increase the entropy weight to enforce binarization [8]; however, Figure 1.1b illustrates why this approach can fail. Specifically, overly strong entropy regularization can

saturate scores at 0 or 1 *without* reliably aligning them with truly important edges, resulting in over-regularized masks. Taken together, Figures 1.1a and 1.1b highlight an inherent trade-off in regularization-only binarization: weak entropy leaves a gray mass near interior values, whereas strong entropy leads to miscalibrated saturation.

## Chapter 4. Proposed Method: BEEP

We introduce BEEP, which leverages self-guidance to address the limited ability of GIB-based models for binarization. As shown in Figure 4.1, the process begins with pseudo-label assignment from the soft scores generated by the base explainer, followed by graph propagation to exploit structural information. These pseudo-labels guide a lightweight explainer in a semi-supervised way, where only confident edges provide explicit supervision. By iteratively updating the labels and explainer over rounds, the score distribution becomes increasingly polarized and bimodal, yielding clearer separation between important and unimportant edges.

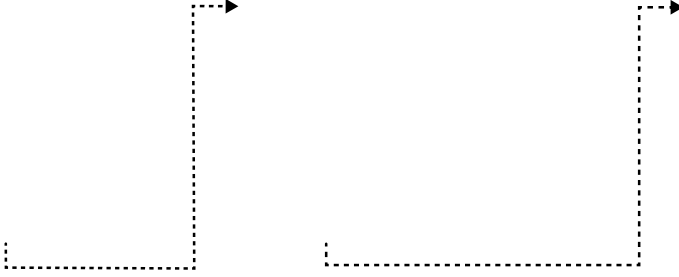


Figure 4.1: Overall workflow of proposed BEEP. Edges falling within the confidence zones (orange or blue regions) are converted into pseudo-labels. The information from these pseudo-labeled edges is then propagated across the graph to provide supervision for unlabeled edges. After one round, pseudo-labeling is repeated, and the guided explainer is fine-tuned with more reliable supervision. Then the score distribution becomes more interpretable, showing clear bimodality and polarization.

### 4.1 Pseudo-labeling via Confidence Zone

Given a graph, BEEP begins by obtaining mask scores from the base explainer, where each edge is assigned a continuous value. We aim to assign pseudo-labels using these mask scores, but they may be inaccurate depending on the performance of the base explainer [14]. To acquire confident pseudo-labels for self-guidance, we divide the scores into *confidence zone* and *gray zone* based on the thresholds statistically earned from the distributions of scores. Pseudo-labels are then assigned only to the confidence zone, either a positive or a negative label, leaving the gray zone.

The interval between the two thresholds defines the gray zone. Edges with scores outside this interval belong to the confidence zone, which is further divided into the positive and negative zones, referred to as the upper tail  $L_+$  and the lower tail  $L_-$ , respectively. This process yields a new dataset in which only edges with high-confidence predictions are assigned pseudo-labels derived from the base explainer. To prevent the guided explainer from becoming over-confident for potentially noisy labels, edges in the gray zone are explicitly ignored when calculating the loss.

Deciding the score thresholds is an important part of our pseudo-labeling. Given the score distribution on the whole dataset from the base explainer  $h^{(b)}$ , i.e.,  $P_s = \{\{s \mid s \in h^{(b)}(G), G \in \mathcal{G}\}\}$ , where

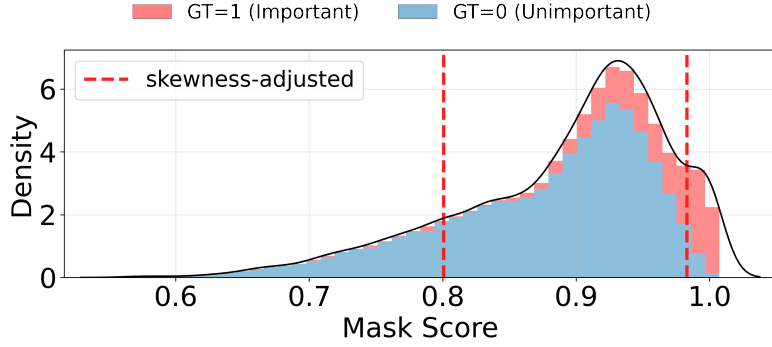


Figure 4.2: Visualization of the thresholding on Mutagenicity. The red and blue regions indicate the proportions of edges with ground-truth labels 1 and 0, respectively, within each score interval. By accounting for skewness in the distribution, our strategy avoids noisy label assignments near ambiguous mid-range regions, yielding refined pseudo-labels.

$\{\{\cdot\}\}$  denotes a multi-set. Here,  $h^{(b)}(G)$  outputs a continuous score matrix, and all its individual entries  $s$  collectively form the distribution. We then compute the skewness  $g_1$  (i.e., Fisher’s moment coefficient of skewness) [12, 11] as follows:

$$g_1 = \mathbb{E}_{s \sim P_s} [(s - \mu)^3] / \sigma^3, \quad (4.1)$$

where  $\mu = \mathbb{E}_s[s]$  and  $\sigma^2 = \mathbb{E}_s[(s - \mu)^2]$ . The positive skewness  $g_1$  indicates a longer upper tail  $L_+$  (positive zone), while the negative value indicates a longer lower tail  $L_-$  (negative zone).

We then determine asymmetric quantile levels to minimize noise and improve label quality, using a base cutoff  $\alpha_0 \in (0, 0.5)$  and a skewness adjustment factor  $c > 0$ :

$$\alpha_{\text{pos}} = \alpha_0 (1 + c \cdot \max(g_1, 0)), \quad (4.2)$$

$$\alpha_{\text{neg}} = \alpha_0 (1 + c \cdot \max(-g_1, 0)). \quad (4.3)$$

Finally, the lower and upper thresholds are defined as follows:

$$t_L = Q_{\alpha_{\text{neg}}}(P_s), \quad t_U = Q_{1-\alpha_{\text{pos}}}(P_s), \quad (4.4)$$

where  $Q_\beta(P_s)$  denotes the  $\beta$ -th quantile of the distribution  $P_s$ . In our pseudo-labeling rule, edges with scores below the lower threshold  $t_L$  are assigned label 0, those above the upper threshold  $t_U$  are assigned label 1, and edges with scores in between remain unlabeled. This thresholding strategy effectively converts the soft importance scores into discrete supervisory signals, enabling the base explainer to guide the subsequent explainer under semi-supervision.

One may alternatively determine thresholds using fixed symmetric quantiles or a simple mean-based rule. However, both approaches overlook potential imbalance in the score distribution and are therefore prone to producing noisy labels. Specifically, the quantile-based method may place cutoffs in noisy mid-ranges when the distribution is skewed, while the mean-based rule implicitly assumes Gaussian-like symmetry that rarely holds in practice. As shown in Figure 4.2, our skewness-adjusted thresholding strategy explicitly accounts for asymmetric tail behavior, resulting in more accurate pseudo-labels and more balanced error compositions.

## 4.2 Training the Guided Explainer

The pseudo-labeling creates a new dataset  $\mathcal{G}' = \{(G, \mathbf{S}', \mathbf{M})\}$ , where  $\mathbf{S}' \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{V}|}$  denotes the pseudo-labels and  $\mathbf{M} \in \{0, 1\}^{|\mathcal{V}| \times |\mathcal{V}|}$  represents the supervision mask;  $m_{uv} = 1$  means that the pseudo-label is given for edge  $(u, v)$ . Our goal is to use the pseudo-labels as stable guidance for training a guided explainer  $h^{(g)}$ , propagating high-confidence signals to all other edges and addressing the limitations of GIB-based models.

**Model Architecture.** Most explainer models, including all baselines used in our experiments, use an MLP backbone. This is natural because the explainer takes as input the node embeddings  $\mathbf{Z}$  produced by a GNN  $f$ ; a lightweight architecture is sufficient, as the graph structure is already encoded in  $\mathbf{Z}$ . In contrast,  $h^{(g)}$  is trained from the pseudo-labels and does not depend on  $\mathbf{Z}$ . We therefore employ a GNN backbone for  $h^{(g)}$ , allowing the semi-supervised signals to propagate effectively across the graph. This architecture allows us to comprehensively capture structural information and flexibly adapt to the topological properties of the input graph, thereby producing task-specific representations for edge-level reasoning.

By default, we use a two-layer GCN followed by an edge classifier as  $h^{(g)}$ . Given  $G = (\mathbf{X}, \mathbf{A})$ , the node embeddings  $\mathbf{H}$  are computed as  $\mathbf{H} = \sigma(\hat{\mathbf{A}} \sigma(\hat{\mathbf{A}} \mathbf{X} \mathbf{W}_1) \mathbf{W}_2)$ , where  $\hat{\mathbf{A}} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$  is the symmetrically normalized adjacency matrix,  $\mathbf{W}_1$  and  $\mathbf{W}_2$  are learnable weights, and  $\sigma(\cdot)$  is the ReLU function. For each edge  $(u, v)$ , we extract the concatenated node embeddings and pass it through a linear edge classifier to obtain the final edge importance:

$$\hat{s}_{uv}^{(g)} = \sigma([\mathbf{h}_u \| \mathbf{h}_v] \mathbf{W}_3 + \mathbf{b}), \quad (4.5)$$

where  $\mathbf{W}_3$  and  $\mathbf{b}$  are learnable parameters.

**Training.** The cross-entropy loss is typically used for binary classification tasks. However, in our case, the fraction of edges that are considered important can vary significantly depending on the dataset. In many cases, the overwhelming majority are easy negatives whose presence scarcely affects the graph label. We thus adopt a *positive class-weighted* binary cross-entropy (BCE) loss, which makes the positive edges contribute more strongly during training, promoting more balanced training:

$$\mathcal{L}_g(h^{(g)}) = \sum_{(u,v) \in \mathcal{E}} \left[ w^+ s'_{uv} \log \hat{s}_{uv}^{(g)} + (1 - s'_{uv}) \log(1 - \hat{s}_{uv}^{(g)}) \right], \quad (4.6)$$

where  $\mathcal{E}$  is the set of edges in  $G$ ,  $s'_{uv} \in \{0, 1\}$  is the pseudo-label for edge  $(u, v)$ , and  $w^+$  is the weight assigned to positive edges to compensate for their scarcity.

## 4.3 Why BEEP Succeeds in Binarization

In Section 3, we have theoretically shown that the GIB objective function fails to create binarized score distributions. We now analyze why the semi-supervised training of the guided explainer in BEEP drives the edge-score distribution toward clear binarization.

To analyze the gradient flow, let  $z_{uv}$  denote the pre-sigmoid logit of the guided edge score for edge  $(u, v)$ , such that  $\hat{s}_{uv}^{(g)} = \sigma(z_{uv})$ . Let  $l_{uv}$  denote the per-edge weighted binary cross-entropy loss from Eq. (4.6), i.e.,  $\mathcal{L}_g = \sum_{(u,v)} l_{uv}$ . Given this objective function, we first theoretically show that the gradient forces the logits in the confidence zones (defined in Section 4.1) to move away from the decision boundary. Importantly, we account for the *tail noise rate*  $\varepsilon_{\pm}$ , which represents the proportion

of incorrect pseudo-labels present in these confident regions. As long as the pseudo-labeling is better than random guessing ( $\varepsilon_{\pm} < 1/2$ ), the expected gradient robustly pushes the scores towards the true binary extremes, effectively polarizing the distribution.

**Lemma 4.1** (Binarization in the confidence zone). *Let the gradient of the per-edge loss  $l_{uv}$  with respect to the logit  $z_{uv}$  be proportional to  $\hat{s}_{uv}^{(g)} - s'_{uv}$ . Assume the tail noise rates satisfy  $\varepsilon_{\pm} < 1/2$ . Conditioned on the positive confidence zone (upper tail)  $L_+$ , the expected gradient is strictly negative, i.e.,  $\mathbb{E}[\partial l_{uv}/\partial z_{uv} \mid L_+] < 0$ , causing the score to polarize toward  $\hat{s}_{uv}^{(g)} \rightarrow 1$ . Conversely, on the negative confidence zone (lower tail)  $L_-$ , the expected gradient is strictly positive, i.e.,  $\mathbb{E}[\partial l_{uv}/\partial z_{uv} \mid L_-] > 0$ , causing  $\hat{s}_{uv}^{(g)} \rightarrow 0$ .*

*Proof Sketch.* The assigned pseudo-label  $s'_{uv}$  is constant within each tail: 1 for  $L_+$  and 0 for  $L_-$ . For the upper tail  $L_+$ , the conditional expectation of the BCE gradient is  $\mathbb{E}[\hat{s}_{uv}^{(g)} - 1 \mid L_+] = \mathbb{E}[\hat{s}_{uv}^{(g)} \mid L_+] - 1$ . Since the sigmoid output  $\hat{s}_{uv}^{(g)}$  is strictly less than 1 (unless trivially converged), this expected gradient is strictly negative. In gradient descent, a negative gradient causes the logit  $z_{uv}$  to increase on average, pushing  $\hat{s}_{uv}^{(g)}$  towards 1. A symmetric argument applies to the lower tail  $L_-$ , where the expected gradient  $\mathbb{E}[\hat{s}_{uv}^{(g)} - 0 \mid L_-] > 0$  causes the logit to decrease, pushing the score towards 0. Details are provided in Appendix A.1.2.  $\square$

Lemma 4.1 explains the direct binarization of pseudo-labeled edges in the confidence zones. The GNN architecture of the guided explainer propagates these binarizing signals to unlabeled edges in the gray zone. This propagation via shared weights and message passing is formalized below.

**Corollary 4.2** (Binarization in the gray zone). *Let  $(u, v)$  be an unlabeled edge in the gray zone. The change in its score,  $\Delta \hat{s}_{uv}^{(g)}$ , after one gradient step with learning rate  $\eta$ , is determined by the alignment of its gradient with the gradients of labeled edges in its neighborhood. Formally,*

$$\Delta \hat{s}_{uv}^{(g)} \approx -\eta \sum_{(i,j) \in L} \frac{\partial l_{ij}}{\partial \hat{s}_{ij}^{(g)}} \langle \nabla_{\theta} \hat{s}_{uv}^{(g)}, \nabla_{\theta} \hat{s}_{ij}^{(g)} \rangle, \quad (4.7)$$

where  $L = L_+ \cup L_-$  is the set of edges in the confidence zones, and the inner product is large and positive for structurally proximate edges. Consequently, the score of an unlabeled edge is pushed toward the dominant pseudo-label of its labeled neighbors.

Additional derivations and analysis are provided in Appendix A.1.3.

*Theoretical Intuition.* The first-order Taylor expansion of the score update is  $\Delta \hat{s}_{uv}^{(g)} \approx -\eta \langle \nabla_{\theta} \hat{s}_{uv}^{(g)}, \nabla_{\theta} \mathcal{L}_g \rangle$ . Decomposing the loss gradient  $\nabla_{\theta} \mathcal{L}_g$  into contributions from individual labeled edges yields the sum above. The update is governed by two factors: the error signal  $\partial l_{ij}/\partial \hat{s}_{ij}^{(g)}$  (which is negative for  $L^+$  and positive for  $L^-$ ) and the topological gradient alignment  $\langle \nabla_{\theta} \hat{s}_{uv}^{(g)}, \nabla_{\theta} \hat{s}_{ij}^{(g)} \rangle$ . Because the GNN shares parameters  $\theta$  across local computational graphs, this alignment inner product is significantly large only when edges  $(u, v)$  and  $(i, j)$  are structurally proximate. Thus, if an unlabeled edge’s neighborhood is dominated by positive pseudo-labels ( $L^+$ ), the sum is dominated by negative error signals multiplied by positive alignments, resulting in  $\Delta \hat{s}_{uv}^{(g)} > 0$ . The converse holds for negatively dominated neighborhoods. In essence, the GNN diffuses binarizing signals from confident edges into the gray zone.  $\square$

Lemma 4.1 and Corollary 4.2 explain why BEEP is able to achieve strong binarization. On confident edges, scores are directly pushed toward 0 or 1. These signals are then propagated through the GNN explainer to unlabeled edges, pushing their scores toward the consensus of their local neighborhood. Consequently, (i) scores move away from the decision boundary, 0.5, across the entire graph; (ii) re-scoring edges after each training stage shifts more edges into the tails and fewer remain in the gray

zone; and (iii) repeating the self-guided loop (relabeling  $\rightarrow$  retrain) through rounds amplifies this effect, making the edge-score distribution increasingly binary.

## 4.4 Linear Time Complexity of BEEP

The time complexity of BEEP is linear with respect to graph size, as formally stated in Lemma 4.3. The running time of BEEP on real-world datasets is also reported in Section 5.6.

**Lemma 4.3** (Time complexity of BEEP). *Let  $\mathcal{D} = \{G_g = (V_g, E_g)\}$  be the dataset,  $N = \sum_g |V_g|$  be the total number of nodes, and  $M = \sum_g |E_g|$  be the number of edges. Assume the guided explainer is an  $L_g$ -layer message-passing GNN (hidden dimension  $h$ ) with an edge head of per-edge cost  $C_e$ , trained for  $T$  epochs per round, and quantiles are computed by sorting. Then, a single round of BEEP over  $\mathcal{D}$  costs*

$$\mathcal{O}\left((T+1)\{L_g(Mh + Nh^2) + MC_e\} + M \log M\right).$$

In each self-guided round, BEEP consists of the following three steps: (i) re-scoring all edges using the current guided explainer; (ii) pseudo-labeling via the confidence zone; and (iii) training the guided explainer with binary cross-entropy (BCE) loss for  $T$  epochs. We analyze the time complexity of each component below.

**(i) Re-scoring.** A single forward pass of an  $L_g$ -layer message-passing GNN over the dataset incurs a cost of

$$\mathcal{O}(L_g(Mh + Nh^2)),$$

where the  $Mh$  term corresponds to neighborhood aggregation over edges and the  $Nh^2$  term accounts for per-node linear transformations. Computing edge importance scores for all edges using the edge scoring head adds an additional cost of  $\mathcal{O}(MC_e)$ . Therefore, the total cost of the re-scoring step is

$$\mathcal{O}(L_g(Mh + Nh^2) + MC_e).$$

**(ii) Pseudo-labeling.** Given the edge scores, computing summary statistics and assigning pseudo-labels can be done with a single linear scan, which costs  $\mathcal{O}(M)$ . The computation of the lower and upper quantile thresholds ( $t_L, t_U$ ) requires sorting all  $M$  scores, resulting in a cost of  $\mathcal{O}(M \log M)$ . This sorting step dominates the procedure, yielding an overall complexity of  $\mathcal{O}(M \log M)$ .

**(iii) Training.** One training epoch of the guided explainer has the same asymptotic cost as one forward and backward pass, namely

$$\mathcal{O}(L_g(Mh + Nh^2) + MC_e).$$

Training the model for  $T$  epochs therefore costs

$$\mathcal{O}\left(T\{L_g(Mh + Nh^2) + MC_e\}\right).$$

**Conclusion.** By summing the costs of steps (i)–(iii), the total time complexity of a single self-guided round is

$$\mathcal{O}\left((T+1)\{L_g(Mh + Nh^2) + MC_e\} + M \log M\right),$$

which establishes the claim in Lemma 4.3.

## Chapter 5. Experiments

### 5.1 Experimental Settings

**Baselines.** We compare BEEP with previous explainer methods: PGExplainer [21], GSAT [22], Mixup-Explainer [43], ProxyExplainer [4] and ConfExplainer [42]. These baselines are post-hoc explainers built upon the graph information bottleneck (GIB) principle, aiming to generate subgraphs that preserve the original GNN prediction. In addition to these GIB-based methods, we also include non-GIB, training-free approaches such as EiG-Search [18] and GOAt [19], which provide architecture-agnostic explanations without learning auxiliary models. These methods produce identical results across repeated runs as they are training-free and deterministic.

**Datasets.** We run experiments on two types of graph datasets: synthetic and real-world graphs. For synthetic graphs, we use the BA-3motifs [31] and MNIST-75sp datasets [23]. For real-world data, we include two molecular datasets, Mutagenicity [5] and Fluoride-Carbonyl [25]. Since graph classification datasets with edge-level ground-truth are highly scarce, we carefully select these to cover both domains. The details of the datasets and the corresponding GNN  $f$  for each dataset are provided in Table 5.1.

BA-3motifs contains synthetic graphs created by attaching motif structures to BA graphs, where the inserted motif edges are treated as ground-truth explanations. Mutagenicity is a molecular classification dataset with  $\text{NH}_2$  and  $\text{NO}_2$  substructures as ground-truth, and we use only the samples with edge-level annotations. Fluoride-Carbonyl is a molecular dataset where fluoride and carbonyl functional groups correspond to ground-truth explanations, and similarly, we use only the samples with edge-level annotations. MNIST-75sp represents digit images as superpixel graphs, where stroke-related edges are treated as ground-truth explanations. For efficiency, we randomly sample 10% of the MNIST-75sp dataset.

**Evaluation Metrics.** We report a comprehensive set of metrics that capture both predictive correctness and distributional interpretability. For predictive performance, we evaluate AUC and F1 against the baselines. Beyond predictive accuracy, we evaluate the shape of the score distributions using bimodality and binarization metrics [24]. To the best of our knowledge, our work is the first attempt to use these distributional metrics in the context of GNN explainers, highlighting how our method goes beyond accuracy.

The degree of bimodality is defined as

$$\text{Bimodality}(p_1, \dots, p_n) = \frac{3}{2} \left( \frac{\gamma^2 + 1}{\kappa} - \frac{1}{3} \right),$$

where  $\gamma = \mu_3/\mu_2^{3/2}$ ,  $\kappa = \mu_4/\mu_2^2$ , and  $\mu_k = \frac{1}{n} \sum_{i=1}^n (p_i - \bar{p})^k$ . This normalized bimodality coefficient becomes closer to 1 if the distribution exhibits clearer separation between modes. We further introduce

Table 5.1: Dataset statistics and target GNN performance.

| Metric       | BA-3motifs | Mutagenicity | Fluoride-Carbonyl | MNIST-75sp |
|--------------|------------|--------------|-------------------|------------|
| GNN Type     | GIN        | GCN          | GCN               | GCN        |
| Val Acc.     | 0.9750     | 0.8651       | 0.9069            | 0.8333     |
| Test Acc.    | 0.9725     | 0.8462       | 0.9002            | 0.8333     |
| Total Graphs | 3000       | 1356         | 8671              | 7988       |
| Avg. Nodes   | 21.92      | 27.57        | 21.36             | 70.55      |
| Avg. Edges   | 14.52      | 28.37        | 22.68             | 295.13     |
| Avg. Degree  | 1.31       | 2.05         | 2.12              | 8.29       |
| Density      | 0.07       | 0.09         | 0.11              | 0.12       |
| Edge-GT Rate | 0.65       | 0.10         | 0.04              | 0.26       |



Table 5.2: Performance comparison between base explainers and BEEP across different datasets in terms of four evaluation metrics. BEEP largely improves predictive metrics (AUC, F1) as well as distributional metrics (binarization and bimodality) over a wide range of explainers. The improvements are consistently observed not only on GIB-based explainers but also on non-GIB methods, demonstrating the general applicability of the proposed guided self-supervision framework.

| Base explainer | Dataset           | AUC                  |                      | Binarization score   |                      | Bimodality           |                      | F1                   |                      |
|----------------|-------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                |                   | Base                 | Guided               | Base                 | Guided               | Base                 | Guided               | Base                 | Guided               |
| PGExplainer    | BA-3motifs        | 0.930 ± 0.025        | <b>0.936 ± 0.031</b> | 0.934 ± 0.028        | <b>0.940 ± 0.026</b> | <b>0.954 ± 0.022</b> | 0.949 ± 0.023        | 0.882 ± 0.013        | <b>0.899 ± 0.018</b> |
|                | Mutagenicity      | 0.855 ± 0.070        | <b>0.968 ± 0.028</b> | 0.245 ± 0.144        | <b>0.426 ± 0.287</b> | 0.314 ± 0.035        | <b>0.531 ± 0.233</b> | 0.278 ± 0.005        | <b>0.376 ± 0.066</b> |
|                | MNIST-75sp        | 0.729 ± 0.001        | <b>0.876 ± 0.074</b> | 0.172 ± 0.330        | <b>0.742 ± 0.154</b> | 0.157 ± 0.053        | <b>0.709 ± 0.067</b> | 0.498 ± 0.001        | <b>0.634 ± 0.077</b> |
|                | Fluoride-Carbonyl | 0.853 ± 0.004        | <b>0.895 ± 0.011</b> | 0.000 ± 0.000        | <b>0.799 ± 0.094</b> | 0.101 ± 0.079        | <b>0.614 ± 0.180</b> | 0.562 ± 0.005        | <b>0.625 ± 0.022</b> |
| GSAT           | BA-3motifs        | 0.937 ± 0.008        | <b>0.957 ± 0.003</b> | 0.540 ± 0.070        | <b>0.956 ± 0.006</b> | 0.775 ± 0.090        | <b>0.966 ± 0.005</b> | 0.903 ± 0.009        | <b>0.925 ± 0.003</b> |
|                | Mutagenicity      | 0.853 ± 0.037        | <b>0.993 ± 0.006</b> | 0.065 ± 0.027        | <b>0.571 ± 0.323</b> | 0.162 ± 0.089        | <b>0.714 ± 0.173</b> | 0.322 ± 0.018        | <b>0.443 ± 0.056</b> |
|                | MNIST-75sp        | 0.730 ± 0.014        | <b>0.890 ± 0.020</b> | 0.178 ± 0.050        | <b>0.754 ± 0.166</b> | 0.182 ± 0.048        | <b>0.787 ± 0.129</b> | 0.526 ± 0.008        | <b>0.675 ± 0.023</b> |
|                | Fluoride-Carbonyl | 0.846 ± 0.018        | <b>0.882 ± 0.017</b> | 0.807 ± 0.096        | <b>0.973 ± 0.004</b> | 0.398 ± 0.061        | <b>0.979 ± 0.002</b> | 0.455 ± 0.013        | <b>0.624 ± 0.039</b> |
| MixupExplainer | BA-3motifs        | <b>0.965 ± 0.002</b> | 0.961 ± 0.003        | 0.766 ± 0.086        | <b>0.928 ± 0.008</b> | 0.788 ± 0.075        | <b>0.937 ± 0.009</b> | <b>0.932 ± 0.008</b> | 0.919 ± 0.012        |
|                | Mutagenicity      | 0.843 ± 0.016        | <b>0.974 ± 0.026</b> | 0.058 ± 0.043        | <b>0.591 ± 0.355</b> | 0.453 ± 0.011        | <b>0.524 ± 0.227</b> | 0.259 ± 0.003        | <b>0.389 ± 0.050</b> |
|                | MNIST-75sp        | 0.741 ± 0.001        | <b>0.900 ± 0.007</b> | 0.110 ± 0.009        | <b>0.839 ± 0.037</b> | 0.130 ± 0.003        | <b>0.847 ± 0.033</b> | 0.506 ± 0.001        | <b>0.664 ± 0.010</b> |
|                | Fluoride-Carbonyl | <b>0.855 ± 0.002</b> | 0.838 ± 0.021        | 0.846 ± 0.020        | <b>0.848 ± 0.055</b> | 0.883 ± 0.010        | <b>0.922 ± 0.026</b> | <b>0.653 ± 0.012</b> | 0.556 ± 0.023        |
| ProxyExplainer | BA-3motifs        | 0.814 ± 0.036        | <b>0.948 ± 0.009</b> | 0.690 ± 0.022        | <b>0.960 ± 0.010</b> | 0.674 ± 0.037        | <b>0.968 ± 0.008</b> | 0.597 ± 0.017        | <b>0.925 ± 0.003</b> |
|                | Mutagenicity      | 0.835 ± 0.078        | <b>0.991 ± 0.008</b> | 0.011 ± 0.012        | <b>0.560 ± 0.351</b> | 0.463 ± 0.040        | <b>0.564 ± 0.274</b> | 0.260 ± 0.002        | <b>0.443 ± 0.142</b> |
|                | MNIST-75sp        | 0.729 ± 0.001        | <b>0.878 ± 0.070</b> | 0.186 ± 0.386        | <b>0.772 ± 0.146</b> | 0.175 ± 0.094        | <b>0.734 ± 0.068</b> | 0.498 ± 0.001        | <b>0.634 ± 0.082</b> |
|                | Fluoride-Carbonyl | 0.802 ± 0.001        | <b>0.845 ± 0.014</b> | 0.896 ± 0.007        | <b>0.899 ± 0.048</b> | 0.690 ± 0.005        | <b>0.947 ± 0.044</b> | <b>0.569 ± 0.001</b> | 0.567 ± 0.013        |
| ConfExplainer  | BA-3motifs        | 0.831 ± 0.038        | <b>0.902 ± 0.018</b> | 0.632 ± 0.330        | <b>0.845 ± 0.083</b> | 0.764 ± 0.144        | <b>0.884 ± 0.050</b> | 0.481 ± 0.048        | <b>0.899 ± 0.014</b> |
|                | Mutagenicity      | 0.882 ± 0.082        | <b>0.947 ± 0.064</b> | 0.366 ± 0.494        | <b>0.624 ± 0.160</b> | 0.480 ± 0.229        | <b>0.688 ± 0.129</b> | 0.271 ± 0.011        | <b>0.430 ± 0.029</b> |
|                | MNIST-75sp        | 0.743 ± 0.020        | <b>0.878 ± 0.049</b> | 0.213 ± 0.394        | <b>0.824 ± 0.025</b> | 0.184 ± 0.108        | <b>0.838 ± 0.020</b> | 0.506 ± 0.011        | <b>0.650 ± 0.071</b> |
|                | Fluoride-Carbonyl | 0.741 ± 0.054        | <b>0.895 ± 0.017</b> | 0.262 ± 0.392        | <b>0.817 ± 0.241</b> | 0.193 ± 0.244        | <b>0.617 ± 0.122</b> | 0.458 ± 0.034        | <b>0.614 ± 0.026</b> |
| EiG-Search     | BA-3motifs        | 0.625 ± 0.000        | <b>0.910 ± 0.005</b> | 0.715 ± 0.000        | <b>0.874 ± 0.043</b> | 0.697 ± 0.000        | <b>0.904 ± 0.031</b> | 0.598 ± 0.000        | <b>0.909 ± 0.003</b> |
|                | Mutagenicity      | 0.504 ± 0.000        | <b>0.953 ± 0.044</b> | 0.682 ± 0.000        | <b>0.758 ± 0.075</b> | 0.705 ± 0.000        | <b>0.754 ± 0.048</b> | 0.157 ± 0.000        | <b>0.495 ± 0.116</b> |
|                | MNIST-75sp        | 0.519 ± 0.000        | <b>0.760 ± 0.072</b> | <b>0.736 ± 0.000</b> | 0.680 ± 0.323        | 0.728 ± 0.000        | <b>0.759 ± 0.252</b> | 0.306 ± 0.000        | <b>0.496 ± 0.030</b> |
|                | Fluoride-Carbonyl | 0.789 ± 0.000        | <b>0.864 ± 0.049</b> | 0.770 ± 0.000        | <b>0.943 ± 0.016</b> | 0.779 ± 0.000        | <b>0.951 ± 0.016</b> | 0.545 ± 0.000        | <b>0.606 ± 0.059</b> |
| GOAt           | BA-3motifs        | 0.771 ± 0.000        | <b>0.919 ± 0.002</b> | 0.422 ± 0.000        | <b>0.937 ± 0.021</b> | 0.363 ± 0.000        | <b>0.943 ± 0.022</b> | 0.683 ± 0.000        | <b>0.890 ± 0.020</b> |
|                | Mutagenicity      | 0.663 ± 0.000        | <b>0.984 ± 0.015</b> | 0.522 ± 0.000        | <b>0.593 ± 0.327</b> | 0.542 ± 0.000        | <b>0.746 ± 0.174</b> | 0.214 ± 0.000        | <b>0.505 ± 0.066</b> |
|                | MNIST-75sp        | 0.504 ± 0.000        | <b>0.908 ± 0.006</b> | <b>0.804 ± 0.000</b> | 0.734 ± 0.091        | 0.516 ± 0.000        | <b>0.781 ± 0.069</b> | 0.417 ± 0.000        | <b>0.708 ± 0.011</b> |
|                | Fluoride-Carbonyl | 0.623 ± 0.000        | <b>0.762 ± 0.013</b> | 0.888 ± 0.000        | <b>0.943 ± 0.013</b> | 0.836 ± 0.000        | <b>0.951 ± 0.012</b> | 0.415 ± 0.000        | <b>0.514 ± 0.015</b> |

the binarization score, which directly quantifies how strongly the scores polarize toward the two extremes.

$$\text{Binarization score}(p_1, \dots, p_n) = \frac{1}{n} \sum_{i=1}^n (2p_i - 1)^2.$$

This measures the concentration of scores toward the binary extremes  $\{0, 1\}$ . Values close to 1 reflect strongly polarized distributions, while values near 0 indicate scores clustered around 0.5.

**Implementation Details.** We adopt each baseline explainer as the base explainer for BEEP. For evaluation, we report the results after the second round of self-guidance across all metrics. Each experiment is repeated over 5 random seeds, and we report the mean and standard deviation. The main hyperparameters,  $\alpha_0$ ,  $c$ , and  $w^+$ , are selected via grid search, where  $\alpha_0$  controls the initial guidance strength,  $c$  adjusts the polarization sharpness, and  $w^+$  balances positive-edge supervision. We use a GCN as the guided explainer architecture for BA-3motifs, Mutagenicity, and MNIST-75sp, while a GIN is used for Fluoride-Carbonyl. We perform grid search over  $\alpha_0 \in \{0.05, 0.075, 0.10, 0.125, 0.15, 0.20\}$ ,  $c \in \{0.25, 0.50, 1.00, 1.50, 2.00, 5.00\}$ , and  $w^+ \in \{0.50, 1.00, 3.00, 5.00\}$ .

## 5.2 Performance Improvement

Table 5.2 includes the AUC performance of BEEP with the baselines. When applied to GIB-based explainers, BEEP generally achieves higher AUC scores than the corresponding base explainers, demonstrating that the guided self-supervision improves the quality of edge-level explanations. The

performance gains are evident on the real-world benchmarks (Mutagenicity, Fluoride-Carbonyl) as well as on synthetic data, indicating that BEEP is effective even when edge labels are scarce and noise is present. Among the datasets, MNIST-75sp is characterized by significantly higher edge density compared to other datasets, presenting a particularly challenging scenario with densely connected structures and noisy edge patterns. The notable success of BEEP on MNIST-75sp highlights its robustness to complex, high-density graphs.

Unlike AUC, the F1 score requires a decision threshold on the predicted edge scores; it thus evaluates the explainer’s ability to make concrete binary decisions by balancing precision and recall. We adopt the mean score as a natural and unbiased threshold, thereby avoiding dataset-specific tuning and providing a fair criterion across different methods. Across most cases, applying BEEP to the base explainer yields comparable or improved F1 scores. These results suggest that BEEP helps the guided explainer better distinguish positive and negative edges in practical decision-making scenarios while maintaining stable binary prediction performance.

Furthermore, BEEP shows strong performance improvements when applied to non-GIB-based explainers, improving AUC and F1 scores. These results suggest that the proposed self-supervision framework is effective beyond specific explainer objectives and can generalize well to more diverse explainer settings.

### 5.3 Analysis of Score Distributions

We perform qualitative and quantitative analysis on the score distributions created by BEEP (*i*) to evaluate whether the resulting score distributions are sufficiently polarized and bimodal and (*ii*) to assess the interpretability proposed by the guided explainer. Our goal is to construct a human-interpretable score distribution; in real-world scenarios, users must rely on the raw score distribution to determine an appropriate threshold for extracting a discrete subgraph due to the lack of ground-truth edge labels.

Figure 5.2 compares the edge-score distributions produced by the base explainer and our guided explainer over self-guided rounds on four datasets. For the base explainer, the scores are often concentrated in a narrow intermediate region or show substantial overlap between label-0 and label-1 edges. This indicates that the base explainer assigns ambiguous scores to many edges, making it difficult to select a reliable threshold for extracting explanatory subgraphs. After applying BEEP, the score distributions become more clearly separated. Edges with label 0 are increasingly pushed toward low scores, whereas edges with label 1 are shifted toward high scores. This trend is already visible after the first round and becomes more pronounced in the second round, where the distributions are more polarized toward the two extremes. As a result, score distributions become progressively more polarized and interpretable, and the associated performance improvements become evident.

As Table 5.2 shows, applying BEEP generally improves both binarization and bimodality quantitatively. The improvements are especially notable on difficult real-world datasets. For example, on Mutagenicity, PGExplainer improves its bimodality score from 0.314 to 0.531, while GSAT increases from 0.162 to 0.714 after applying BEEP. On MNIST-75sp, where densely connected structures make explanation particularly challenging, substantial gains are consistently observed across most explainers. For instance, MixupExplainer improves its binarization score from 0.110 to 0.839, and ConfExplainer improves its bimodality score from 0.184 to 0.838. These results indicate that the proposed guided framework remains effective even under noisy and high-density graph structures.

## 5.4 Ablation Studies

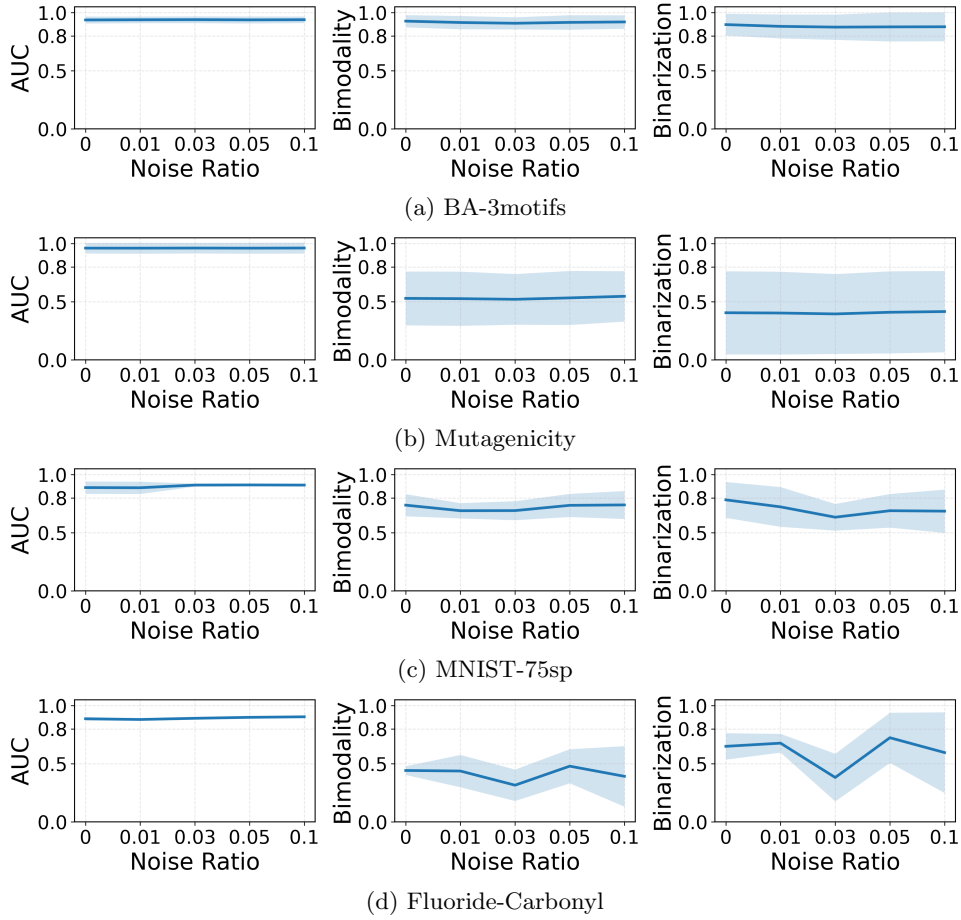


Figure 5.1: Effect of injected pseudo-label noise on explanation quality. Explanation accuracy remains stable as noise increases, while distributional metrics degrade only mildly, indicating that BEEP is robust to imperfect pseudo-labels.

We perform two types of ablation studies considering BEEP as a modular framework: *(i)* We investigate the sensitivity of BEEP to pseudo-label quality by perturbing the confidence scores used during training, highlighting its limited dependence on highly accurate pseudo-labels. *(ii)* We examine how performance evolves as the number of rounds increases, showing if our module converges stably and yields progressively more interpretable explanations.

**Pseudo-label Reliability.** To evaluate the stability of BEEP under noisy confidence scores, we conduct a noise-injection study. In this setting, a fixed ratio of pseudo-labels is flipped to the opposite label at each round, while all hyperparameters remain identical to those in the main experiments. As shown in Figure 5.1, AUC remains largely unchanged even when the noise ratio increases, indicating that its performance is robust to perturbations in the pseudo-labels. The bimodality and binarization change only slightly across noise levels and remain high. Interestingly, in the Fluoride-Carbonyl case, a small amount of noise occasionally leads to more polarized and stable score distributions, suggesting that the injected perturbations may regularize the inherent noise already present in the pseudo-labels. Overall, these observations indicate that BEEP does not rely heavily on perfectly clean pseudo-labels.

**Round-wise Improvements.** We further study how the performance of BEEP evolves as the number of

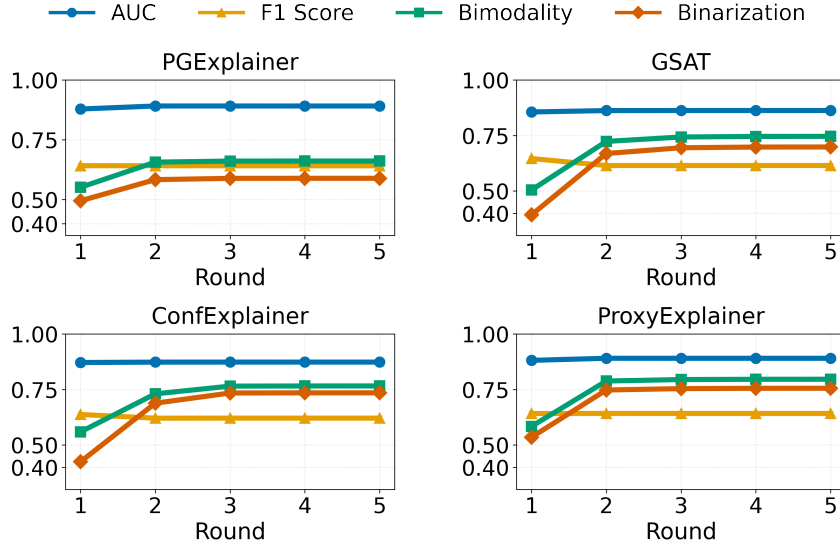


Figure 5.2: Change of various metrics over multiple rounds of BEEP on MNIST-75sp. As rounds proceed, AUC tends to remain similar, while distributional metrics such as bimodality and binarization score steadily rise and converge.

rounds increases beyond 2. In Figure 5.2, while the AUC shows little change beyond slight improvements, the binarization score and bimodality continue to rise and eventually converge, revealing progressively sharper distinctions between positive and negative edges. For F1 scores, the highest values typically appear in the initial round, after which they decrease slightly but eventually stabilize, reflecting a trade-off where sharper and more interpretable score distributions are achieved at the cost of marginal predictive balance. Nevertheless, the stabilized F1 scores remain sufficiently high compared to the overall results in Table 5.2, indicating that the improved interpretability is achieved without substantial degradation in explanation quality. These results demonstrate that our multi-round self-guidance converges stably and enhances interpretability over successive rounds.

## 5.5 Robustness to Hyperparameter Choices

We examine the stability and sensitivity of BEEP with respect to the three main hyperparameters,  $\alpha_0$ ,  $c$ , and  $w^+$  introduced in Section 5.1. The hyperparameters were manually selected to achieve balanced improvements across both predictive and distributional metrics, rather than being chosen solely based on the highest value of a single metric. Figure 5.3 visualizes the hyperparameter sensitivity using heatmaps based on the interquartile range of the performance distributions across all hyperparameter combinations. Darker colors indicate more stable performance, while brighter colors represent larger variability. Overall, the predictive metrics (AUC and F1) exhibit small IQR values across most datasets and base explainers, indicating that the predictive quality of the generated explanations changes only moderately under different hyperparameter choices. Meanwhile, since the distributional metrics (binarization and bimodality) are directly affected by how aggressively the confidence zones polarize the score distributions, these metrics tend to show slightly higher sensitivity than the predictive metrics. In particular,  $\alpha_0$  and  $c$  determine the size and asymmetry of the confidence zones, while  $w^+$  controls the strength of positive-edge supervision during guided training. As a result, different hyperparameter configurations can produce noticeably different degrees of score polarization even when the predictive sensitivity re-

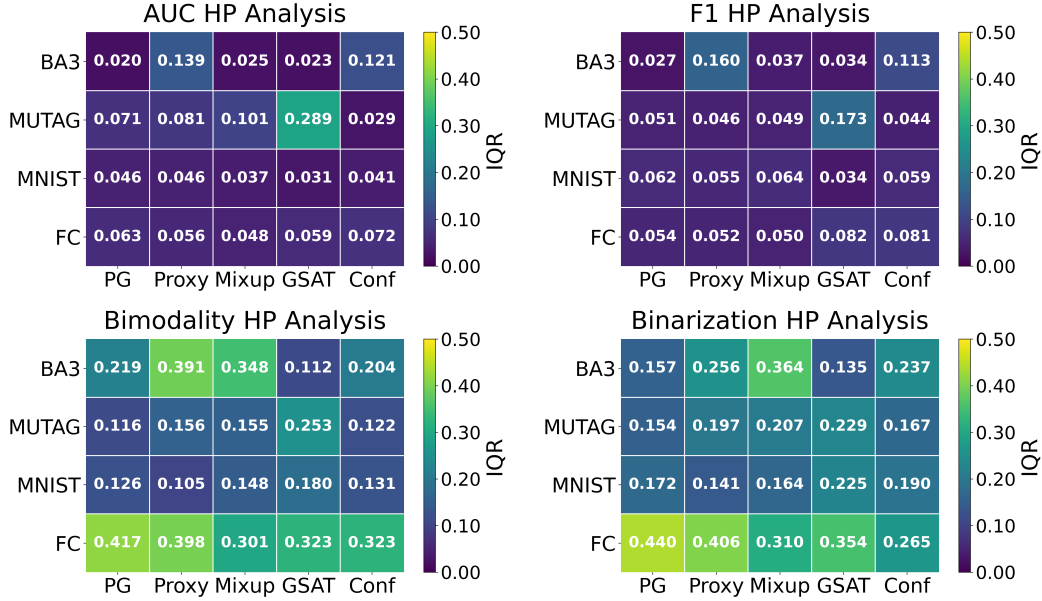


Figure 5.3: Hyperparameter sensitivity analysis summarized with heatmaps. The  $x$ -axis and  $y$ -axis labels denote base explainers and datasets, respectively. They show the performance trends under different combinations of  $\alpha_0$ ,  $c$ , and  $w^+$  in terms of AUC, bimodality, binarization, and F1 score.

Table 5.3: Training and inference time (in seconds) across datasets, comparing the computational cost of base explainers and their guided variants. Guided training time includes the multi-round training cost (R1+R2) and pseudo-label generation time, while guided inference time corresponds to the final Round2 explainer.

| Base Explainer | Training Time (s) |        |        |        |        |        |         |        | Inference Time (s) |        |       |        |       |        |       |        |
|----------------|-------------------|--------|--------|--------|--------|--------|---------|--------|--------------------|--------|-------|--------|-------|--------|-------|--------|
|                | BA3               |        | MUTAG  |        | MNIST  |        | FC      |        | BA3                |        | MUTAG |        | MNIST |        | FC    |        |
|                | Base              | Guided | Base   | Guided | Base   | Guided | Base    | Guided | Base               | Guided | Base  | Guided | Base  | Guided | Base  | Guided |
| PGExplainer    | 14.33             | 14.49  | 11.10  | 5.60   | 47.69  | 45.42  | 45.07   | 3.76   | 0.112              | 0.036  | 0.054 | 0.020  | 0.555 | 0.446  | 0.050 | 0.023  |
| GSAT           | 15.51             | 14.97  | 9.12   | 6.88   | 69.02  | 55.74  | 88.06   | 6.94   | 0.086              | 0.037  | 0.057 | 0.020  | 0.533 | 0.458  | 0.055 | 0.022  |
| MixupExplainer | 222.20            | 12.24  | 124.17 | 5.91   | 808.31 | 56.23  | 1230.81 | 3.66   | 0.114              | 0.035  | 0.054 | 0.022  | 0.544 | 0.448  | 0.049 | 0.022  |
| ProxyExplainer | 152.48            | 17.30  | 108.76 | 6.45   | 859.47 | 49.63  | 877.94  | 3.98   | 0.120              | 0.035  | 0.052 | 0.021  | 0.547 | 0.471  | 0.048 | 0.023  |
| ConfExplainer  | 24.00             | 8.77   | 29.48  | 6.97   | 74.51  | 58.13  | 91.17   | 3.75   | 0.116              | 0.038  | 0.057 | 0.021  | 0.549 | 0.450  | 0.048 | 0.022  |

mains small. Several Fluoride-Carbonyl cases exhibit relatively larger IQR values for the distributional metrics. This behavior is likely related to the structural characteristics of Fluoride-Carbonyl, where the explanatory substructures are sparse and embedded within complex molecular graphs, making the polarization process more sensitive to the confidence-zone configuration. Although several cases exhibit relatively large IQR values, particularly for the distributional metrics, the observed variability does not overturn the overall improvement trends achieved by applying BEEP as shown in Table 5.2.

## 5.6 Computational Cost

We measure the training time of BEEP to assess its computational cost. To ensure a fair comparison, the same experimental settings, including the number of epochs, validation procedure, and early stopping strategy, were consistently applied. Training time is measured end-to-end and includes validation performed every epoch. For BEEP, it includes both rounds (1 and 2) as well as the intermediate pseudo-labeling steps. For models whose training goes beyond 30 epochs, we apply early stopping with a patience of 30 to ensure flexibility across datasets.

Table 5.3 shows the additional training time taken by the multi-round refinement of BEEP. The guided explainer is trained on top of a pre-trained base explainer, and the extra cost added by additional rounds remains relatively small across datasets. Importantly, even on MNIST-75sp, our largest and densest dataset, BEEP maintains cost-effectiveness. This indicates that our method scales well with increasing graph size and density, suggesting that scalability is not a limiting factor in practical deployment. It is notable that, consistent with its cost-effective training, BEEP also exhibits efficient inference, introducing no noticeable overhead compared to existing baselines.

## 5.7 Additional Qualitative Results

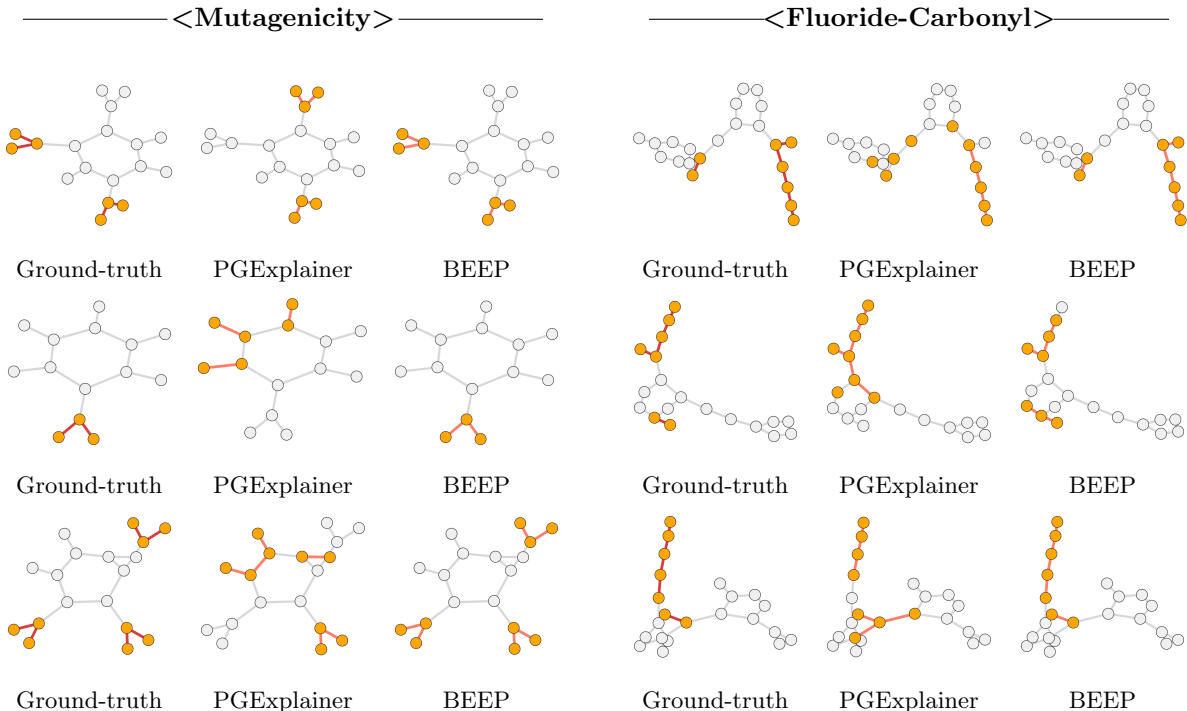


Figure 5.4: Qualitative comparison of explanation results on Mutagenicity and Fluoride-Carbonyl. Each row corresponds to a different molecular graph and compares the ground-truth explanatory subgraph, the explanation generated by the base explainer (PGExplainer), and the explanation refined by BEEP.

We provide additional qualitative results to visually assess whether BEEP can accurately identify explanatory subgraphs on real-world molecular datasets. Figure 5.4 presents the ground-truth subgraphs, the explanations generated by the base explainer (PGExplainer), and the explanations refined by BEEP.

On Mutagenicity, the baseline explainer often captures only a partial region of the functional group or includes irrelevant surrounding structures. In contrast, BEEP more accurately isolates the ground-truth-related subgraphs through the proposed self-guided refinement process, producing explanations that are both clearer and more complete. On Fluoride-Carbonyl, the task is considerably more challenging due to the complex molecular structures and ambiguous edge patterns. Although BEEP still does not perfectly recover the entire important substructure in some cases, it consistently generates explanations that are more aligned with the ground-truth patterns compared to the baseline explainer. These qualitative results demonstrate that the proposed guided framework improves the structural fidelity and interpretability of the extracted explanatory subgraphs.

## Chapter 6. Conclusion

In this work, we presented BEEP to enhance post-hoc explainers for graph neural networks by driving score distributions toward clearer binarization and bimodality for improved interpretability. BEEP transforms soft edge importance scores into reliable supervisory signals and amplifies them through self-guidance, progressively sharpening edge-level distinctions. We further provided theoretical analysis showing why semi-supervision with pseudo-labeling drives scores away from ambiguous mid-ranges, yielding stronger binarization compared to prior GIB-based approaches. Across both synthetic and real-world datasets, BEEP consistently improves the accuracy and interpretability of existing post-hoc explainers. It also introduces only small additional computational overhead compared to training the original base explainers. Beyond these results, our framework motivates several future directions, including adaptive round selection or threshold-free approaches to pseudo-labeling.

## Chapter A. Appendix

### A.1 Theoretical Insights

#### A.1.1 Derivation of the GIB Stationary Point

The GIB surrogate in Equation (3.1) can be differentiated coordinate-wise as

$$\frac{\partial \mathcal{L}_b}{\partial \hat{s}_{uv}^{(b)}} = \underbrace{\frac{\partial}{\partial \hat{s}_{uv}^{(b)}} \text{CE}(\hat{Y}, \hat{Y}^*(\hat{\mathbf{S}}^{(b)}))}_{g_{uv}(\hat{\mathbf{S}}^{(b)})} + \lambda_1 + \lambda_2 \log \frac{1 - \hat{s}_{uv}^{(b)}}{\hat{s}_{uv}^{(b)}} \quad (\text{A.1})$$

Here  $g_{uv}(\hat{\mathbf{S}}^{(b)}) = \langle \frac{\partial \text{CE}}{\partial z}, \frac{\partial z}{\partial \hat{s}_{uv}^{(b)}} \rangle$  by the chain rule with logit  $z$  of  $f(G^*)$ . When the masked prediction already matches (or nearly matches) the original one, the fidelity gradient satisfies  $\|\frac{\partial \text{CE}}{\partial z}\| \approx 0$ , and often  $\frac{\partial z}{\partial \hat{s}_{uv}^{(b)}} = 0$  for many edges. On  $g_{uv} \approx 0$ , this stationary condition yields a closed-form solution strictly between 0 and 1. Starting from the gradient condition in Equation (A.1), the stationary point satisfies

$$0 = g_{uv} + \lambda_1 + \lambda_2 \log \frac{1 - \hat{s}_{uv}^{(b)}}{\hat{s}_{uv}^{(b)}}.$$

On  $g_{uv} \approx 0$ , this yields

$$s_{uv}^{(g)*} = \frac{1}{1 + \exp(-\lambda_1/\lambda_2)} = \sigma(\lambda_1/\lambda_2) \in (0, 1).$$

That is, each coordinate admits a natural fractional fixed point strictly between 0 and 1.

#### A.1.2 Derivation of Expected Push Direction for Confidence Zone

*Proof of Lemma 4.1.* We provide a formal proof for the two cases stated in the lemma. The core of the argument is that the pseudo-label  $s'_{uv}$  is constant within each tail, which creates a systematic bias in the gradient of the loss function.

**Upper Tail ( $L_+$ ):** We want to show  $\mathbb{E}\left[\frac{\partial \ell}{\partial z_{uv}} \mid (u, v) \in L_+\right] < 0$ .

By definition, the gradient of the binary cross-entropy (BCE) loss  $\ell$  with respect to the edge logit  $z_{uv}$  is given by:

$$\frac{\partial \ell}{\partial z_{uv}} = \hat{s}_{uv}^{(g)} - s'_{uv}$$

where  $\hat{s}_{uv}^{(g)} = \sigma(z_{uv})$  is the output of the sigmoid function.

We take the conditional expectation over all edges  $(u, v)$  belonging to the upper tail  $L_+ = \{(u, v) : \hat{s}_{uv}^{(b)} \geq t_U\}$ :

$$\mathbb{E}\left[\frac{\partial \ell}{\partial z_{uv}} \mid L_+\right] = \mathbb{E}\left[\hat{s}_{uv}^{(g)} - s'_{uv} \mid L_+\right]$$

By the linearity of expectation, this can be separated into two terms:

$$\mathbb{E}\left[\frac{\partial \ell}{\partial z_{uv}} \mid L_+\right] = \mathbb{E}\left[\hat{s}_{uv}^{(g)} \mid L_+\right] - \mathbb{E}[s'_{uv} \mid L_+]$$



According to the pseudo-labeling rule defined in Section 3.1, any edge  $(u, v)$  in the upper confidence zone  $L_+$  is assigned the pseudo-label  $s'_{uv} = 1$ . This value is constant for all members of the set  $L_+$ . Therefore, its expectation is:

$$\mathbb{E}[s'_{uv} \mid L_+] = \mathbb{E}[1 \mid L_+] = 1$$

Substituting this back, we get:

$$\mathbb{E}\left[\frac{\partial \ell}{\partial z_{uv}} \mid L_+\right] = \mathbb{E}[\hat{s}_{uv}^{(g)} \mid L_+] - 1$$

Since the score  $\hat{s}_{uv}^{(g)}$  is the output of a sigmoid function, it is strictly bounded in the interval  $[0, 1]$ . Its expectation over any set of edges must also lie within this interval. Unless the guided explainer has perfectly converged such that  $\hat{s}_{uv}^{(g)} = 1$  for all edges in  $L_+$  (a trivial stationary point), the expectation will be strictly less than 1:

$$\mathbb{E}[\hat{s}_{uv}^{(g)} \mid L_+] < 1$$

It follows that the expected gradient is strictly negative:

$$\mathbb{E}\left[\frac{\partial \ell}{\partial z_{uv}} \mid L_+\right] < 0$$

In a gradient descent update step, the change in the logit is proportional to the negative of the gradient (i.e.,  $\Delta z_{uv} \propto -\frac{\partial \ell}{\partial z_{uv}}$ ). Since the expected gradient is negative, the expected change in the logit is positive, causing  $z_{uv}$  to increase on average and thus pushing the score  $\hat{s}_{uv}^{(g)}$  towards 1.

**Lower Tail ( $L_-$ ):** The argument is symmetric for the lower tail  $L_- = \{(u, v) : \hat{s}_{uv}^{(b)} \leq t_L\}$ . The conditional expectation is:

$$\mathbb{E}\left[\frac{\partial \ell}{\partial z_{uv}} \mid L_-\right] = \mathbb{E}[\hat{s}_{uv}^{(g)} \mid L_-] - \mathbb{E}[s'_{uv} \mid L_-]$$

For any edge in  $L_-$ , the assigned pseudo-label is  $s'_{uv} = 0$ . Thus,  $\mathbb{E}[s'_{uv} \mid L_-] = 0$ . The expression simplifies to:

$$\mathbb{E}\left[\frac{\partial \ell}{\partial z_{uv}} \mid L_-\right] = \mathbb{E}[\hat{s}_{uv}^{(g)} \mid L_-]$$

Unless the model has trivially converged to  $\hat{s}_{uv}^{(g)} = 0$  for all edges in  $L_-$ , its expected score will be strictly greater than 0:

$$\mathbb{E}[\hat{s}_{uv}^{(g)} \mid L_-] > 0$$

Therefore, the expected gradient is strictly positive. This positive gradient causes the logit  $z_{uv}$  to decrease on average during gradient descent, pushing the score  $\hat{s}_{uv}^{(g)}$  towards 0. This completes the proof.  $\square$

### A.1.3 Derivation of Binarization in the Gray Zone via Neighborhood Influence

*Proof of Corollary 4.2.* The corollary states that the score of an unlabeled edge  $(u, v)$  is updated based on the pseudo-labels of its structurally proximate labeled neighbors. We prove this by analyzing the flow of gradients through the shared parameters  $\theta$  of the GNN-based guided explainer.

**Step 1: Deriving the Score Update Equation.** The parameters  $\theta$  of the guided explainer are updated via gradient descent:  $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}_g(\theta_t)$ . The first-order Taylor approximation for the change in the score of an unlabeled edge  $(u, v) \notin L$  is:

$$\begin{aligned} \Delta \hat{s}_{uv}^{(g)} &= \hat{s}_{uv}^{(g)}(\theta_{t+1}) - \hat{s}_{uv}^{(g)}(\theta_t) \\ &\approx \left\langle \nabla_{\theta} \hat{s}_{uv}^{(g)}(\theta_t), \theta_{t+1} - \theta_t \right\rangle \\ &= -\eta \left\langle \nabla_{\theta} \hat{s}_{uv}^{(g)}, \nabla_{\theta} \mathcal{L}_g \right\rangle. \end{aligned} \quad (\text{A.2})$$

The loss  $\mathcal{L}_g$  is a sum over only the labeled edges in  $L$ . Therefore, its gradient is:

$$\nabla_{\theta} \mathcal{L}_g = \sum_{(i,j) \in L} \nabla_{\theta} l_w(\hat{s}_{ij}^{(g)}, s'_{ij}) = \sum_{(i,j) \in L} \frac{\partial l_w}{\partial \hat{s}_{ij}^{(g)}} \nabla_{\theta} \hat{s}_{ij}^{(g)} \quad (\text{A.3})$$

Substituting this decomposition back into the expression for  $\Delta \hat{s}_{uv}^{(g)}$  yields the equation stated in the corollary:

$$\Delta \hat{s}_{uv}^{(g)} \approx -\eta \sum_{(i,j) \in L} \frac{\partial l_w}{\partial \hat{s}_{ij}^{(g)}} \langle \nabla_{\theta} \hat{s}_{uv}^{(g)}, \nabla_{\theta} \hat{s}_{ij}^{(g)} \rangle$$

**Step 2: Analyzing the Components of the Update.** The update is a sum of contributions from all labeled edges. Each contribution is a product of two terms:

(1) **The Error Signal** ( $\partial l_w / \partial \hat{s}_{ij}^{(g)}$ ): This scalar term indicates the direction of correction for a labeled edge  $(i, j)$ . For the weighted binary cross-entropy loss  $l_w(\hat{s}, s') = -[w_+ s' \log \hat{s} + (1 - s') \log(1 - \hat{s})]$ , its derivative is:

$$\frac{\partial l_w}{\partial \hat{s}_{ij}^{(g)}} = \begin{cases} -w_+ / \hat{s}_{ij}^{(g)} < 0 & \text{if } (i, j) \in L^+ \text{ (i.e., } s'_{ij} = 1) \\ 1 / (1 - \hat{s}_{ij}^{(g)}) > 0 & \text{if } (i, j) \in L^- \text{ (i.e., } s'_{ij} = 0) \end{cases}$$

The signal is negative for positive pseudo-labels (pushing the score up) and positive for negative pseudo-labels (pushing the score down).

(2) **The Gradient Alignment** ( $\langle \nabla_{\theta} \hat{s}_{uv}^{(g)}, \nabla_{\theta} \hat{s}_{ij}^{(g)} \rangle$ ): This inner product measures the influence of a learning signal from  $(i, j)$  on the score of  $(u, v)$ . The score  $\hat{s}_{uv}^{(g)}$  is a function of the final node embeddings, which are themselves complex functions of the shared GNN parameters  $\theta$  and the graph topology. The gradient  $\nabla_{\theta} \hat{s}_{uv}^{(g)}$  reflects the sensitivity of the score to changes in these shared parameters. This alignment term is large and positive if and only if the computational graphs for  $(u, v)$  and  $(i, j)$  have significant overlap. In a GNN, this occurs when edges are structurally proximate (e.g., are incident to the same node or are 1-hop neighbors). The shared weights  $\theta$  in the message-passing layers are updated based on local neighborhood structures, so nearby edges will have highly correlated gradients with respect to  $\theta$ . Conversely, for distant edges, the alignment is near zero.

**Step 3: Synthesis of Neighborhood Influence.** We analyze the net effect on  $\Delta \hat{s}_{uv}^{(g)}$  based on the dominant pseudo-label in the neighborhood of  $(u, v)$ .

- **Case 1: Positively Dominated Neighborhood.** Assume the labeled edges structurally proximate to  $(u, v)$  are mostly in  $L^+$ . The sum for the update will be dominated by these positive neighbors. For each such neighbor  $(i, j) \in L^+$ , its contribution to the sum is a product of a negative error signal and a large positive alignment term. This product is negative. The total sum is therefore a large negative value. The final change is  $\Delta \hat{s}_{uv}^{(g)} \approx (-\eta) \times (\text{large negative sum}) > 0$ . Thus, the score  $\hat{s}_{uv}^{(g)}$  increases.

- **Case 2: Negatively Dominated Neighborhood.** Assume the labeled neighbors of  $(u, v)$  are mostly in  $L^-$ . For each such neighbor  $(i, j) \in L^-$ , its contribution is a product of a positive error signal and a large positive alignment term. This product is positive. The total sum is a large positive value. The final change is  $\Delta \hat{s}_{uv}^{(g)} \approx (-\eta) \times (\text{large positive sum}) < 0$ . Thus, the score  $\hat{s}_{uv}^{(g)}$  decreases.

In conclusion, the GNN architecture inherently propagates supervision from labeled edges to their unlabeled neighbors through updates to its shared weights. An unlabeled edge's score is pushed towards the consensus pseudo-label of its local neighborhood, providing a formal mechanism for the binarization of scores in the gray zone.  $\square$

#### A.1.4 Assumptions for Theoretical Insights

**A1 (Calibration/monotonicity).** Let  $\eta_{uv} := \Pr(s_{uv} = 1 \mid G)$ . There exists  $\delta \in [0, \frac{1}{2})$  such that

$$\sup_{(u,v) \in \mathcal{E}} |s_{uv}^{(b)} - \eta_{uv}| \leq \delta. \quad (\text{A.4})$$

**A2 (Low-density decision boundary).** There exist  $C > 0$  and  $\kappa > 0$  such that

$$\Pr(|\eta_{uv} - \frac{1}{2}| \leq \gamma) \leq C \gamma^\kappa, \quad \forall \gamma > 0. \quad (\text{A.5})$$

#### A.1.5 Risk and Empirical Objective

Denote by  $L \subseteq \mathcal{E}$  the labeled (confidence zone) edges obtained from the skew-aware quantile rule in Section 4.1, and by  $\mathcal{N}$  an edge-neighborhood system (e.g., line-graph adjacency) used only for analysis.

Let  $\mathcal{D}$  be a distribution over graphs  $G = (\mathbf{X}, \mathbf{A})$  with edge set  $\mathcal{E}$ . For each edge  $(u, v) \in \mathcal{E}$ , the (unknown) target is the binary label  $s_{uv} \in \{0, 1\}$ . Let  $\ell_w$  be the weighted binary cross entropy. We consider the population *tail risk*:

$$R_{\text{tail}}(h^{(g)}) = \mathbb{E}_{G \sim \mathcal{D}} \left[ \frac{1}{|L|} \sum_{(u,v) \in L} \ell(\hat{s}_{uv}^{(g)}, s) \right],$$

which matches our evaluation protocol on confident edges.  $h^{(g)}$  is trained with the *supervised-on-tails* objective:

$$\begin{aligned} \hat{R}_{\text{sup}}(h^{(g)}) &= \frac{1}{|L|} \sum_{e \in L} \ell_w(\hat{s}_{uv}^{(g)}, s'), \\ \ell_w(\hat{s}, s') &= -[w_+ s' \log \hat{s} + (1 - s') \log(1 - \hat{s})]. \end{aligned} \quad (\text{A.6})$$

**Tail purity (noise upper bounds).** Define the tail noise rates  $\varepsilon_+ := \Pr(s = 0 \mid \hat{s}_{uv}^{(b)} \geq t_U)$  and  $\varepsilon_- := \Pr(s = 1 \mid \hat{s}_{uv}^{(b)} \leq t_L)$ , where edge  $(u, v)$  is arbitrarily chosen. Under a mild calibration/monotonicity condition on  $\hat{s}_{uv}^{(b)}$  (Appendix A.1.4), one has  $\varepsilon_+ \leq 1 - t_U + \delta$  and  $\varepsilon_- \leq t_L + \delta$ , so both tails are majority-correct when  $t_U > 1/2 + \delta$  and  $t_L < 1/2 - \delta$ .

**Generalization bound on confident edges.** Let  $\mathcal{H}_g$  denote the hypothesis class of guided explainers and  $\mathfrak{R}_{n_\ell}(\mathcal{H}_g)$  be the empirical Rademacher complexity[45, 46] of  $\mathcal{H}_g$  over  $n_\ell$  labeled edges, where  $n_\ell := |L|$ .

**Theorem A.1** (Generalization of BEEP on the confidence zone). *With probability at least  $1 - \delta$  over the draw of training graphs and the confidence zone selection, every  $h^{(g)} \in \mathcal{H}_g$  satisfies*

$$\begin{aligned}
R_{\text{tail}}(h^{(g)}) \leq & \underbrace{\widehat{R}_{\text{sup}}(h^{(g)})}_{\text{empirical supervised risk}} \\
& + \underbrace{2 \mathfrak{R}_{n_\ell}(\mathcal{H}_g) + c_1 \sqrt{\frac{\log(1/\delta)}{n_\ell}}}_{\text{statistical complexity}} \\
& + \underbrace{c_2(\varepsilon_+ + \varepsilon_-)}_{\text{pseudo-label noise}}. \tag{A.7}
\end{aligned}$$

Here  $c_1, c_2 > 0$  are universal constants.

*Sketch.* Condition on  $L$  and apply uniform convergence for bounded Lipschitz losses to control the first two terms; bound the difference between true labels and pseudo-labels on  $L$  by the tail noise rates. A complete proof is deferred to Appendix A.1.6.

Compared to a base MLP explainer, the GNN structure (message passing, parameter sharing, permutation invariance) reduces  $\mathfrak{R}_{n_\ell}(\mathcal{H}_g)$  at a matched hidden dimension, yielding a tighter statistical term. In practice this translates into better generalization on the confident edges.

### A.1.6 Proof of Theorem A.1

We use the clipped BCE  $\ell_\tau(\hat{s}, y)$  with  $\hat{s} \in [\tau, 1 - \tau]$ ; it is  $B_\tau$ -bounded and  $L_\tau$ -Lipschitz.

**Step 1: Uniform convergence on the selected tails.** Condition on the data-dependent tail set  $L$  (measurable w.r.t. base scores). By symmetrization and contraction, for any  $\delta \in (0, 1)$ , with probability at least  $1 - \delta/2$ ,

$$\begin{aligned}
\mathbb{E} \left[ \frac{1}{n_\ell} \sum_{e \in L} \ell(\hat{s}_{uv}^{(g)}, s'_{uv}) \right] & \leq \frac{1}{n_\ell} \sum_{e \in L} \ell(\hat{s}_{uv}^{(g)}, s'_{uv}) \\
& + 2 \mathfrak{R}_{n_\ell}(\mathcal{H}_g) + c_1 \sqrt{\frac{\log(2/\delta)}{n_\ell}}. \tag{A.8}
\end{aligned}$$

**Step 2: Noise correction on the tails.** On  $L_+ = \{(u, v) : \hat{s}_{uv}^{(b)} \geq t_U\}$  put  $s' = 1$  and on  $L_- = \{(u, v) : \hat{s}_{uv}^{(b)} \leq t_L\}$  put  $y' = 0$ . Since  $\ell_\tau$  is bounded by  $B_\tau$ ,

$$|\ell_\tau(\hat{s}_{uv}^{(g)}, s) - \ell_\tau(\hat{s}_{uv}^{(g)}, s')| \leq B_\tau \mathbf{1}\{s \neq s'\}.$$

Taking expectations and using  $\Pr(s_{uv} \neq s'_{uv} \mid L_+) = \varepsilon_+$  and  $\Pr(s_{uv} \neq s'_{uv} \mid L_-) = \varepsilon_-$  gives

$$\mathbb{E} \left[ \frac{1}{n_\ell} \sum_{(u,v) \in L} \ell(\hat{s}_{uv}^{(g)}, s_{uv}) \right] \leq \mathbb{E} \left[ \frac{1}{n_\ell} \sum_{e \in L} \ell(\hat{s}_{uv}^{(g)}, s'_{uv}) \right] + c_2(\varepsilon_+ + \varepsilon_-).$$

**Step 3: Combine.** Combine Steps 1–2 and remove the clipping constants into  $c_1, c_2$  to obtain (A.7).

### A.1.7 Extension to Full-Edge Risk

If one wishes to upper bound the full-edge risk

$$R(h^{(g)}) = \mathbb{E} \left[ \frac{1}{|\mathcal{E}|} \sum_{(u,v) \in \mathcal{E}} \ell(\hat{s}_{uv}^{(g)}, s) \right], \tag{A.9}$$

a simple coverage term  $B_\tau \frac{|\mathcal{E} \setminus L|}{|\mathcal{E}|}$  can be added (vacuous but assumption-free); alternatively, introducing an unlabeled consistency regularizer (e.g., a Laplacian penalty) during analysis yields

$$R(h^{(g)}) \leq \frac{|L|}{|\mathcal{E}|} R_{\text{tail}}(h^{(g)}) + \lambda \Omega_{\text{Lap}}(h^{(g)}) + c_3 \frac{\Lambda}{|\mathcal{N}|},$$

under a smoothness prior on the target function.

**Remark 1** (No unlabeled regularizer is used in training). We do not use any unlabeled (e.g., Laplacian) term in training. Any  $\Omega_{\text{Lap}}$  that appears in the Appendix is an analysis device for bounding the contribution of unlabeled edges when one extends the bound from  $R_{\text{tail}}$  to the full-edge risk. In all experiments we set  $\lambda=0$  and optimize only (A.6).

Note that our algorithm sets  $\lambda=0$  (Remark 1), and this extension is theoretically sound.

### A.1.8 VC-Based Comparison Between Base and Guided Explainers

We briefly recall the definition of VC dimension ([47, 48]), which provides a principled way to quantify the capacity of a function class and analyze its generalization behavior.

**Definition 1** (VC Dimension). *The Vapnik–Chervonenkis (VC) dimension of a function class  $\mathcal{H}$  is defined as the size of the largest set of points that can be shattered by  $\mathcal{H}$ . A set is said to be shattered if, for every possible binary labeling of the set, there exists a function in  $\mathcal{H}$  that realizes it.*

**Definition 2** (Generalization gap). *Let  $\mathcal{X}$  and  $\mathcal{Y}$  be feature and label spaces, respectively. For a function  $h : \mathcal{X} \rightarrow \mathcal{Y}$ , a bounded loss  $\mathcal{L}^1$  and a dataset  $\{(x_i, y_i)\}_{i=1}^m$ , the generalization gap is*

$$\Delta(h) := \mathbb{E}_{(x,y) \sim \mathcal{P}}[\mathcal{L}(h(x), y)] - \frac{1}{m} \sum_{i=1}^m \mathcal{L}(h(x_i), y_i).$$

**Lemma A.2** (VC-dimension generalization bound). *Let hypothesis class  $\mathcal{H}$  have VC dimension  $d_{\text{VC}}$  and  $\mathcal{L}$  be bounded. Then, for any  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$ , every  $h \in \mathcal{H}$  satisfies*

$$\Delta(h) \leq C \sqrt{\frac{d_{\text{VC}} + \log(1/\delta)}{m}},$$

for a universal constant  $C > 0$ .

**Lemma A.3** (VC-dimension of piecewise-linear MLPs). *Consider an MLP with piecewise-linear activations,  $W$  trainable parameters and  $L$  layers. Its VC dimension satisfies  $d_{\text{VC}} = \mathcal{O}(W L \log W)$ .*

**Lemma A.4** (Structure-aware VC control for GNNs). *For a GNN with ReLU nonlinearities and message-passing with a fixed aggregation operator, the VC dimension is upper-bounded by a function of the final hidden width  $d_{L-1}$  and structural factors. In particular, for matched widths, a GNN typically admits a strictly smaller  $d_{\text{VC}}$  than a plain MLP due to these structural priors.*

**Theorem A.5** (VC-dimension based generalization gaps). *Let  $h^{(b)}$  be the base (MLP) explainer and  $h^{(g)}$  the guided (GNN) explainer. Let  $m$  be the number of training edges used for  $h^{(b)}$ , and  $n_\ell$  the number of pseudo-labeled (confidence-zone) edges used for  $h^{(g)}$ . Then, for any  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$ ,*

$$\Delta(h^{(b)}) \leq \tilde{C} \sqrt{\frac{W L \log W + \log(1/\delta)}{m}}, \quad (\text{A.10})$$

$$\Delta(h^{(g)}) \leq \tilde{C} \sqrt{\frac{d_{L-1} + \log(1/\delta)}{n_\ell}}, \quad (\text{A.11})$$

---

<sup>1</sup>In our setting, the clipped BCE  $\ell_\tau(\hat{s}, y)$  with  $\hat{s} \in [\tau, 1 - \tau]$  is bounded and hence satisfies this assumption.

for a universal constant  $\tilde{C} > 0$ .

**Discussion (Why the guided gap is tighter).** By (A.10)–(A.11), the base MLP gap scales with  $W L \log W$ , while the guided GNN gap is controlled essentially by the last-layer width  $d_{L-1}$ . Because message passing, parameter sharing, and permutation invariance reduce the effective capacity, the guided bound is typically *tighter*, even though it uses only  $n_\ell \leq m$  confident samples. This matches the empirical reliability improvements for BEEP.

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### 연구 업 적

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